

Inference

Grad admissions data

Interested in predicting grad admission from GPA, GRE score, and school rank:

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Suppose we want to estimate the effect of GRE, after accounting for GPA and school rank. How would I compute a confidence interval for β_2 ?

Previously: confidence intervals

Normal confidence interval for β_2 :

$$\hat{\beta}_2 \pm z^* SE_{\hat{\beta}_2}$$

Options for estimating $SE_{\hat{\beta}_2}$:

- + Bootstrap estimate of standard error
- + Estimate of standard error from R output

Confidence interval

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.989979073	1.139950936	-3.500132	0.0004650273
gpa	0.804037549	0.331819298	2.423119	0.0153878974
gre	0.002264426	0.001093998	2.069864	0.0384651284
as.factor(rank)2	-0.675442928	0.316489661	-2.134171	0.0328288188
as.factor(rank)3	-1.340203916	0.345306418	-3.881202	0.0001039415
as.factor(rank)4	-1.551463677	0.417831633	-3.713131	0.0002047107

How do I interpret this confidence interval?

Another question

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

How could we address this research question?

Hypothesis testing

What steps are involved in hypothesis testing?

Specifying the hypotheses

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

H_0 :

H_A :

Confidence intervals and hypothesis testing

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0 : \beta_2 = 0 \quad H_A : \beta_2 \neq 0$$

Recall that we would **reject** H_0 at significance level α if and only if a $(1 - \alpha)100\%$ confidence interval for β_2 **does not** contain 0. (Equivalently, a $(1 - \alpha)100$ confidence interval for e^{β_2} **does not** contain 1).

Computing a p-value

To compute a p-value, we need:

- + a test statistic
- + the distribution of the test statistic **if H_0 were correct**

What test statistic did we use for testing single coefficients in *linear* models?

The null distribution

To compute a p-value, we need the **null distribution** (distribution of the test statistic under H_0).

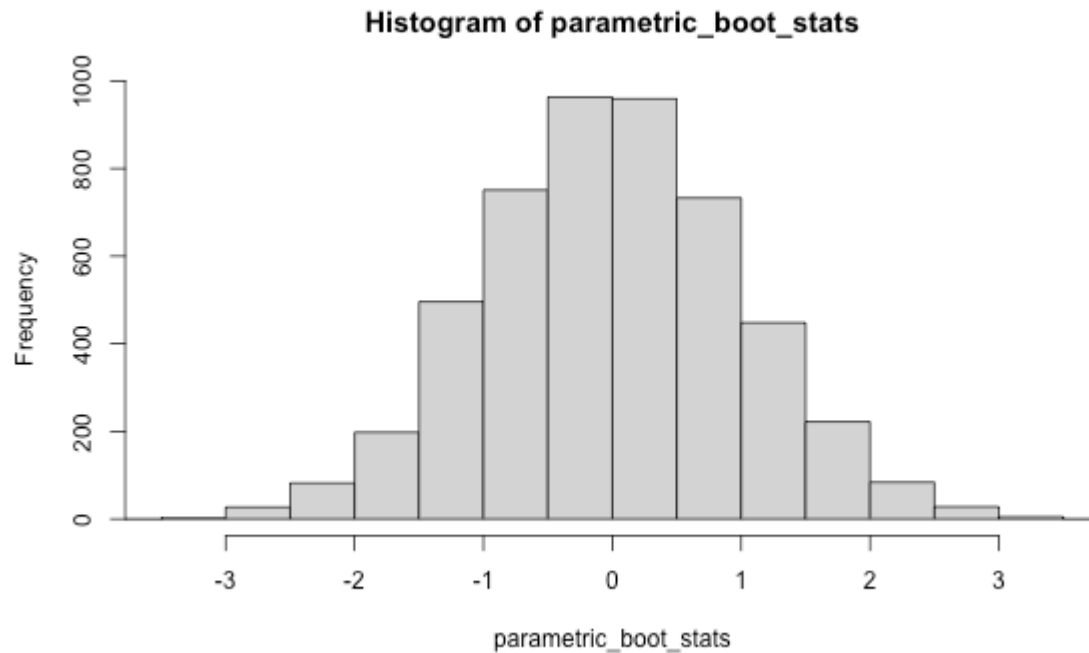
Does *bootstrap resampling* (what we used to create a confidence interval) give us the **null distribution**?

Parametric bootstrapping

Class activity

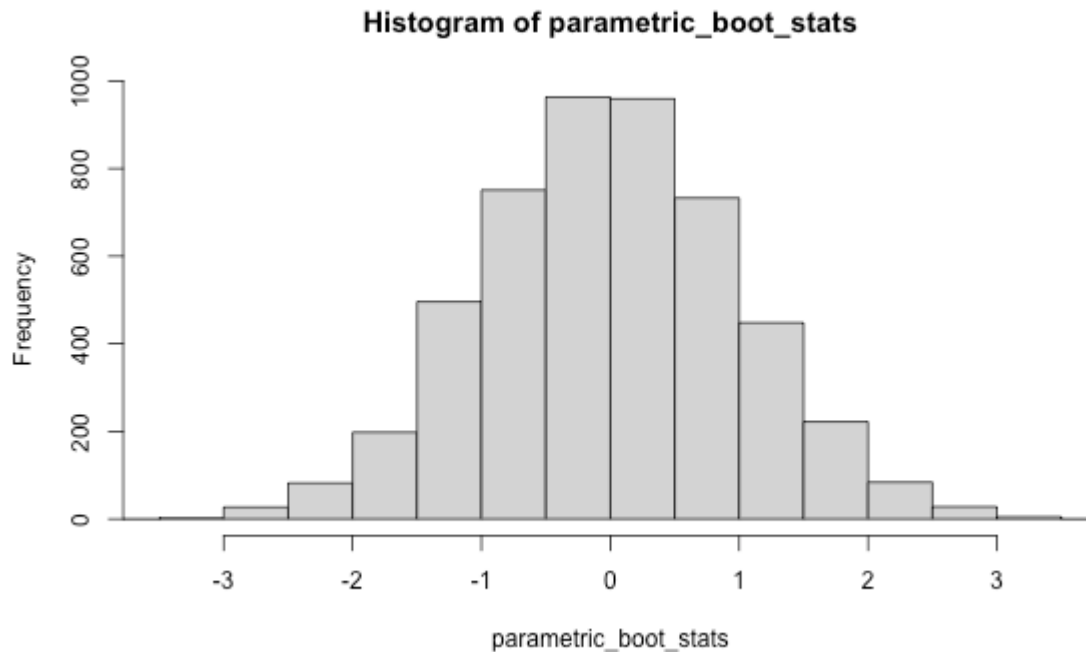
- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

Activity



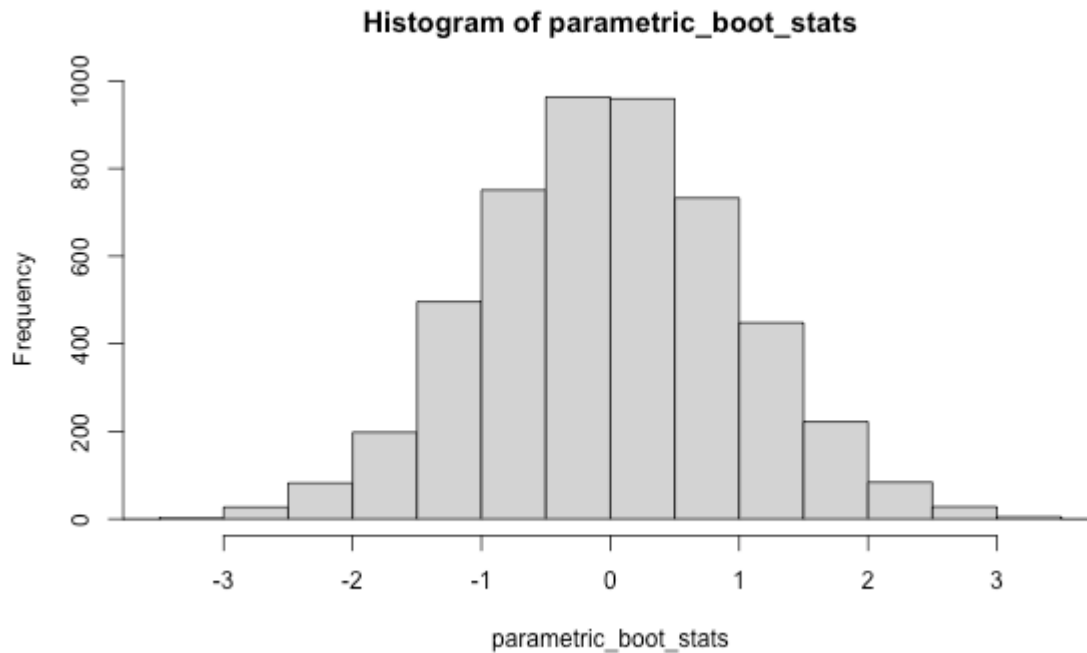
What does this distribution look like?

Activity



How unusual is the observed test statistic of 2.07, compared to this null distribution?

p-value



```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

p-value

```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

```
summary(m_full)$coefficients
```

##		Estimate	Std. Error	z value	Pr(> z)
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